



2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

JUNE 16–20 | UNIVERSITY OF MICHIGAN

Personnel Training, Requirements & Ethics

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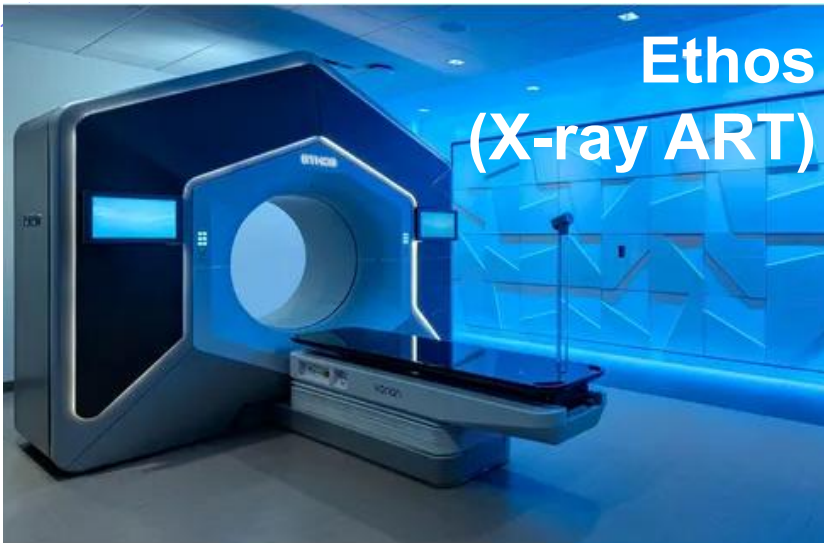
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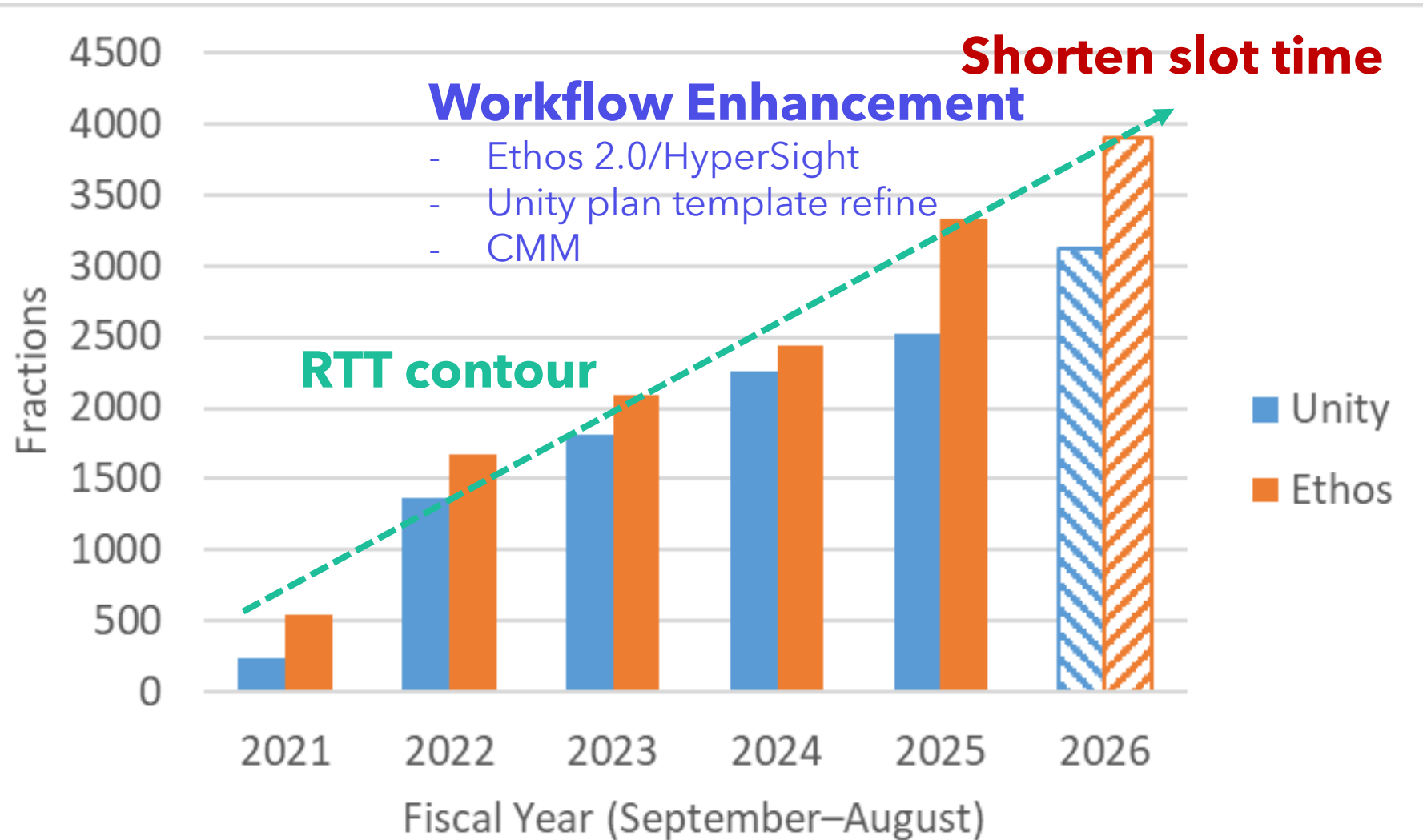
Conflict of Interest Disclosure

- Research support from Varian Medical Systems through a Statement of Work (SOW) agreement.
 - Adapt-On-Demand workflow and tool development
 - Advanced Therapist Training for Ethos Adaptive Therapy
- Course Director, Ethos ART Training Course, University of Texas Southwestern Medical Center.

- New expansion 05/2021: advanced and adaptive RT hub- Adaptive Radiation Oncology Center (AROC)
- 2 Halcyon, 2 Ethos, 2 Unity, 1 RefleXion and 1 Brachy Suite
- CT, MRI, PET/CT



UTSW Adaptive Treatment Volume



From “Can we adapt?” to “We adapt well — every day”

ART has moved beyond its experimental phase to a routine, real-time clinical capability — complex, high-stakes decisions made at the console, under time pressure.

Platforms expand what's possible

CBCT-, MR-, and PET/biology-guided systems each raise the bar for speed, coordination, and modality-specific expertise.

The bottleneck shifts to people

Reliable ART takes more than technical proficiency with devices — it takes coordinated, credentialed teams.

The goal is consistency

Not whether a single case can be adapted, but whether the program adapts safely and well — every single day.

What this talk covers

IN FOCUS — the people side

- ▶ Governance, decision rights & team structure
- ▶ Core ART tasks organized by skillset, not job title
- ▶ Training, credentialing & privileging
- ▶ Ethics within fast, data-rich workflows (Refer to book)
- ▶ Staffing models, hidden workload & physics FTEs



Three adaptive platforms, one team challenge

Same core competencies — different modality-specific demands.

CBCT-gART

e.g., Varian Ethos

- Online, fully **automated**
- Optimization objectives **locked** during session
- Limited soft-tissue contrast
- Preplan robustness is critical
- Status: mature, widely deployed

MR-gART

e.g., Elekta Unity/ViewRay MRIdian

- Online, objectives **editable**
- Electron-return effect to manage
- **Bore / cryostat beam constraints**
- **MR safety screening** required
- Status: mature, widely deployed

BGRT

PET-Linac

- Offline re-planning today
- Driven by the PET / biologic signal
- Tracer timing & uptake window
- **Nuclear-medicine coordination**
- Status: online capability maturing



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Newly Available Systems

- Fan beam CT for planning
- **Flexibility in online objective** adjustment or adapt/not adapt



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If an adaptive platform allows online editing of optimization objectives, what is the most likely staffing implication?

01

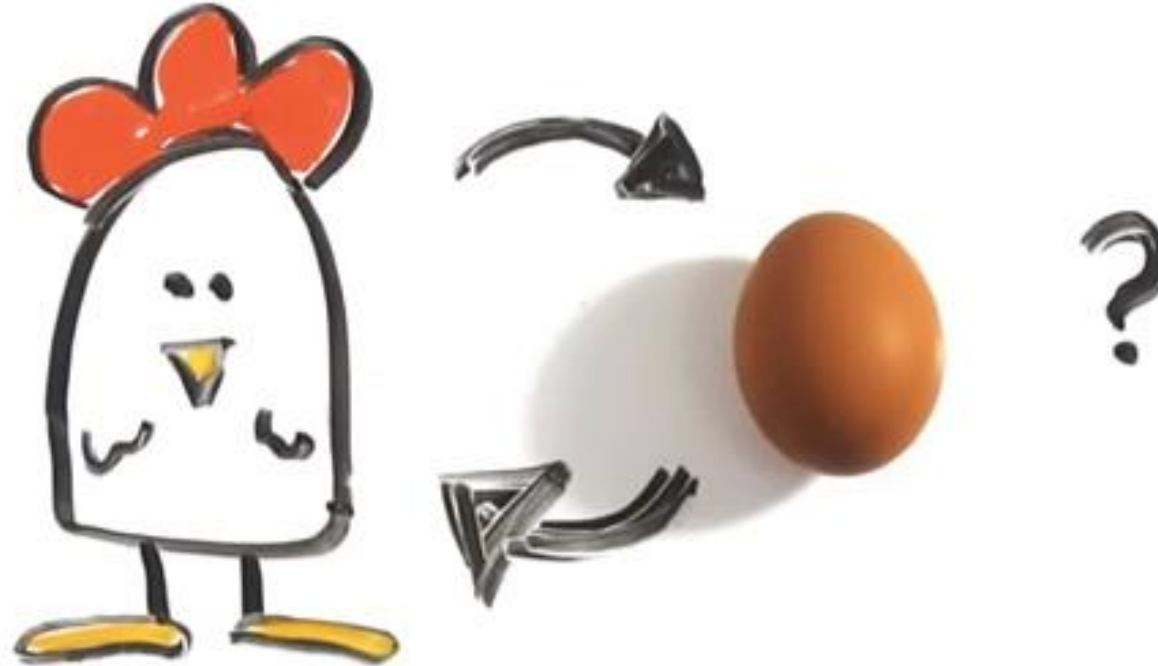
Governance & Team Structure

- The organizational backbone that makes ART safe and sustainable



It's a chicken and egg problem

- Too busy, reimbursement model is unresolved
- Less patients, less motivation to make systematic changes



ART needs structure — a layered operational framework

Adaptive Oversight Committee

Strategy, standards & governance — sets the institutional “playbook”



Disease-Oriented Working Groups

Translate guidelines into site-specific protocols & templates



Daily / Weekly Huddles

Real-time problem solving and continuous micro-improvement



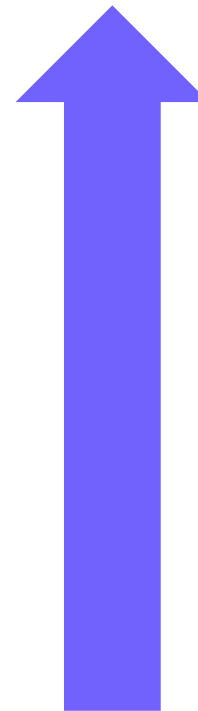
Incident Learning & Data Feedback

Track events, near-misses & vault-time analytics to improve

Starting the
program



Refine and
expand the
program



The Adaptive Oversight Committee — sets the “playbook”

Who: physician leads · physicists · dosimetrists · therapists · administrative stakeholders

1 Strategic direction

Define rollout sequence, patient-selection criteria, and institutional goals for ART.

2 Standardize workflows

Approve end-to-end clinical & technical workflows across CBCT, MR, BGRT and disease sites.

3 Credentialing & privileging

Set training requirements, proficiency benchmarks, and ongoing competency maintenance.

4 Resource advocacy

Assess machine time, personnel & infrastructure needs; communicate them to leadership.

5 Quality oversight

Review safety events, near-misses, operational metrics & outcomes to drive improvement.

Operationalizing: working groups, huddles & learning loops

Together, centralized governance + disease expertise + frontline collaboration + data feedback = a coordinated system of care.

Disease-oriented working groups

- Translate institutional guidelines → site-specific protocols
- Case & peer review to refine candidate-selection criteria
- Build adaptive-intent templates, objectives & checklists
- Bridge clinical services and the adaptive-technology team

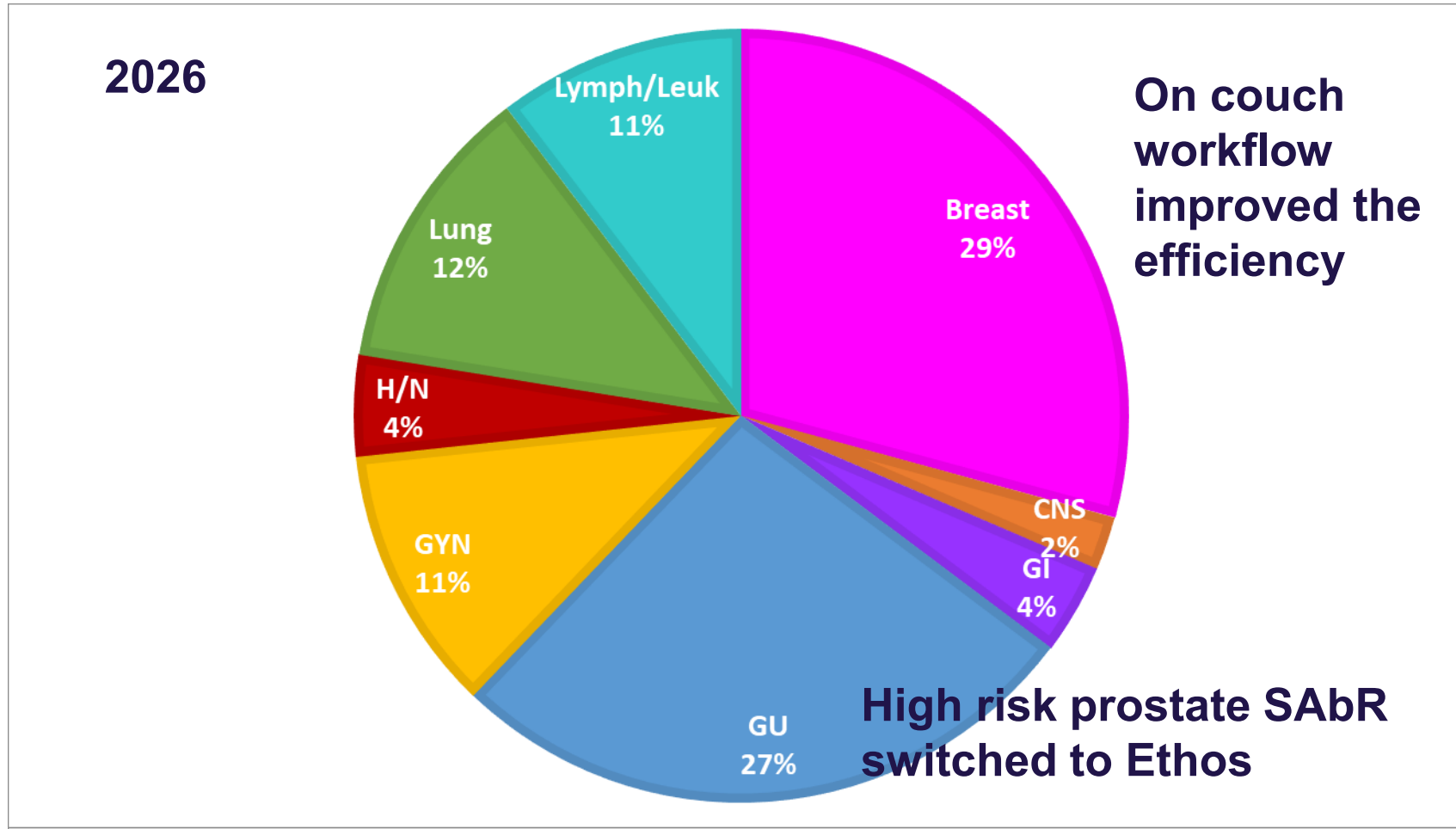
Daily / weekly huddles

- Therapist–Physicist: on-couch issues (setup, motion, image quality)
- Planner–Physicist: hard cases, optimization, QA failures
- Bidirectional feedback — frontline staff escalate & propose fixes

Incident learning & analytics

- Incident Learning System (ILS): safety events, bugs, breakdowns
- Informs daily troubleshooting and committee-level strategy
- Learn from both failures and successes

Example: UTSW Practice Mix Evolution by Disease Site (2022–2026, Ethos)



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02

The Work Itself — Tasks & Skills

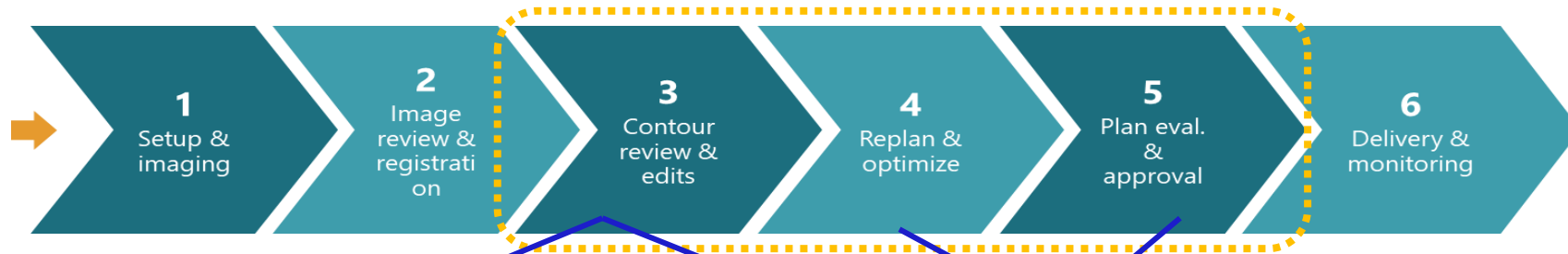
- Organizing ART by the work and the skillset, not the job title
- From the task, think about who is the best person to perform + **how much training gap**
 - as current, physicists are doing the heavy lifting for training (initial & continuous)

The adaptive session: a chain of linked tasks

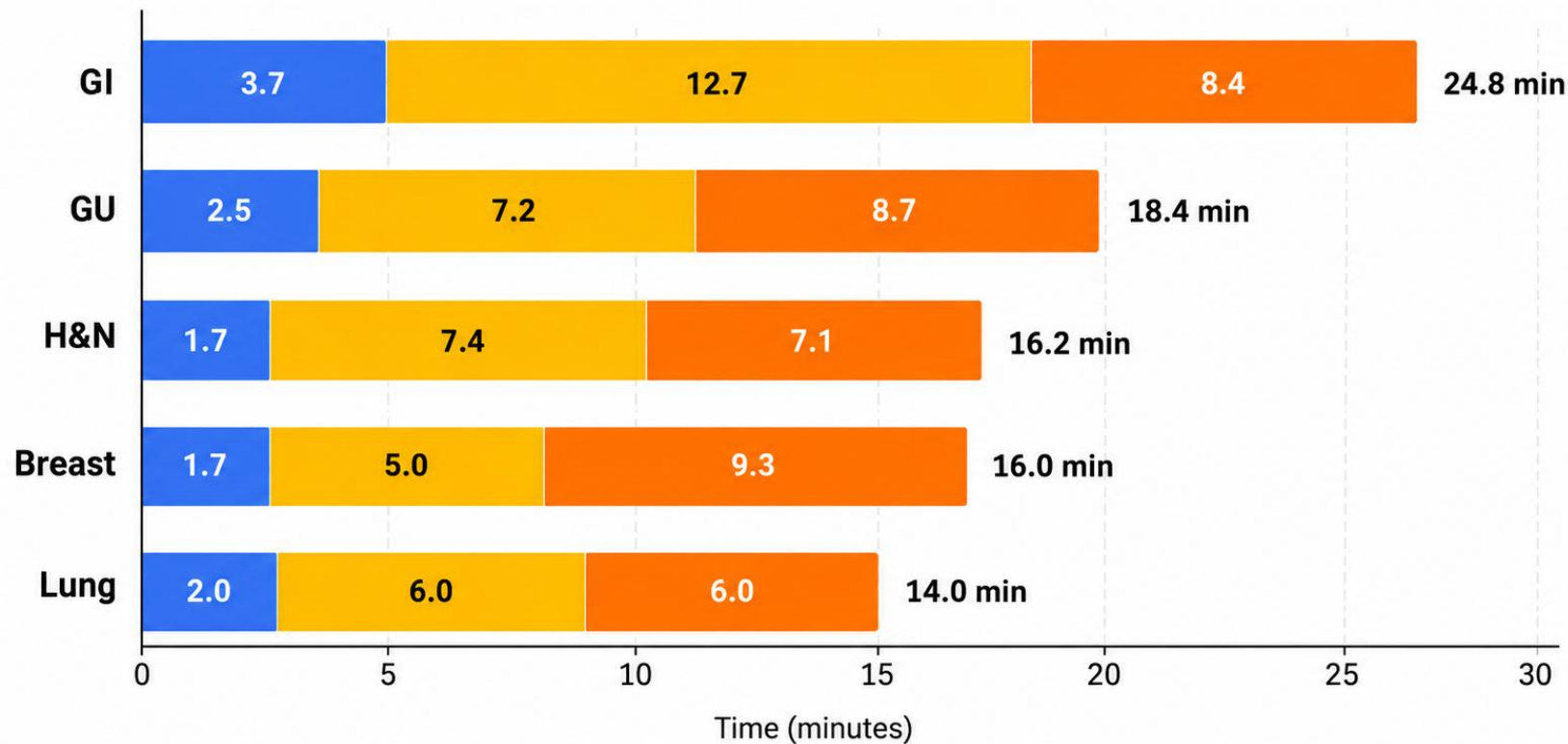


Why the chain matters: a weak link early — e.g., a poor registration — ripples downstream: degraded contour propagation, longer on-table editing, lower plan quality, and compromised clinical intent.

Adaptive workflow time (Ethos Example)



■ Influencer (OAR edits) ■ Target (Delineation) ■ Plan Generation & Review



Credential to the task, not the title

ART is interdisciplinary & time-pressured

Fixed, discipline-bounded roles don't fit the speed and coordination ART demands.

Organize by task and skillset

Describe responsibilities as the work itself — contour, register, replan, approve, deliver, QA.

Many professions, clearly-defined tasks

The same task may be performed by different professionals — if competent, credentialed & available.

Core competency = judgment under pressure

Less about who you are, more about designing robust strategies and deciding well at the console.

Three staffing archetypes — match expertise to availability

Physician-driven \$\$\$\$

- RTT perform setup, image, delivery
- MD performs / directs target & key-OAR edits and gives final approval
- Physicist ensures quality, safety & delivery readiness

Physicist-driven \$\$

- Physicist refines contours (pre-approved rules) & leads replanning
- Physician performs focused review & approval
- Useful in early ramp-up needing technical stewardship

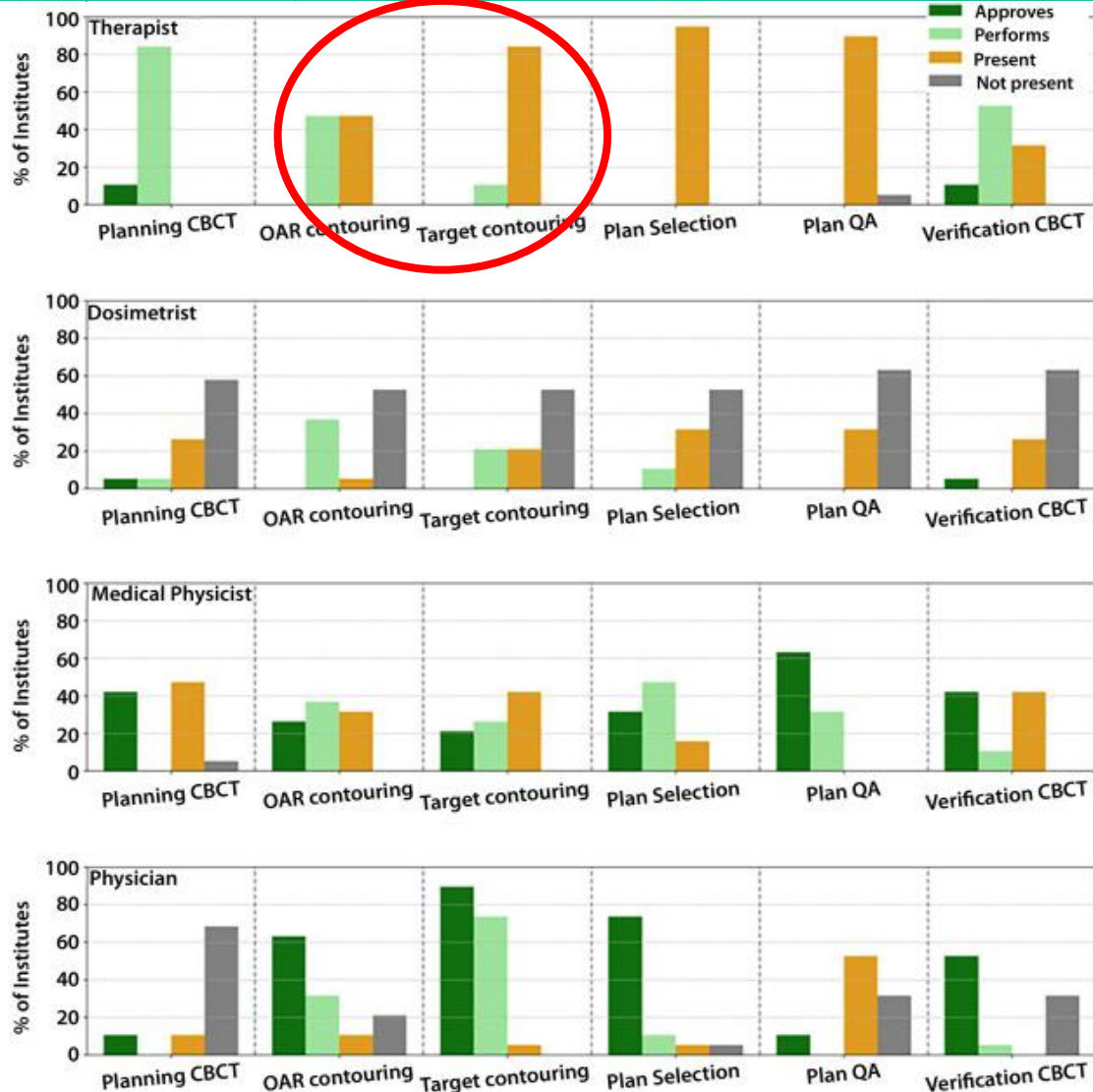
Therapist-enabled \$

- Credentialed RTT refines predefined OARs
- Dosimetrist/physicist leads replanning; physician signs off
- Leverages therapist availability to boost throughput

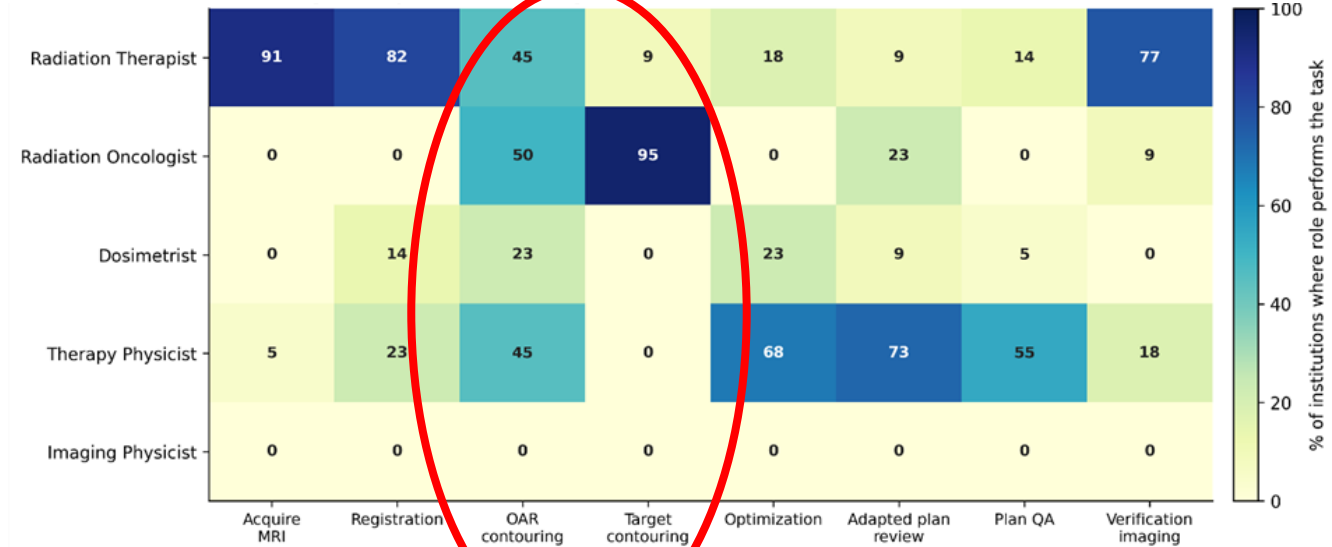
Whatever the mix, operationalize it: (i) clear decision rights · (ii) explicit handoffs at each step · (iii) escalation rules · (iv) minimum in-room presence & backup coverage. The physicist always owns safety & deliverability.

Current Status of On-Couch Staffing (North America Survey)

Ethos



MRgRT (unpublished data)



BRIEF REPORT

Cone Beam Computed Tomography-Guided Online Adaptive Radiation Therapy: Clinical Insights From a Nationwide Staffing Survey

Ti Bai, PhD,^a Xenia Ray, PhD,^b David Parsons, PhD,^a and Mu-Han Lin, PhD^a

^aDepartment of Radiation Oncology, UT Southwestern Medical Center, Dallas, Texas; and ^bDepartment of Radiation Medicine & Applied Sciences, UC San Diego Health, San Diego, California

Received Nov 4, 2024; Revised Feb 13, 2025; Accepted for publication Mar 22, 2025



It starts before the patient is on the couch

Reference plan creation. Contour / verify OARs, derive planning structures, choose the field arrangement, and optimize a deliverable plan with acceptable trade-offs and robustness for on-couch adaptation.

Reference plan review / chart check. A comprehensive evaluation before approval: image quality, contours, margins, isocenters, plan parameters, secondary dose calc — plus all on-treatment dependencies (objectives & priorities, structure naming, DIR/propagation assumptions, verification imaging & action levels).

EXAMPLE OF MODALITY-SPECIFIC DEMANDS → What's New to Learn?

CBCT-gART (Ethos)

Optimization is locked online — the reference plan must be inherently robust to anatomical change.

MR-gART (Unity)

Objectives editable online; account for the electron-return effect and bore / cryostat constraints.

BGRT

Incorporate the PET signal; dose fall-off differs from a conventional linac.



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Setup, imaging & the first big decision

Setup & image acquisition. Reproduce simulation conditions; confirm immobilization, isocenter marks & indexing; verify image quality is sufficient for on-couch contour review; consider contrast to improve lesion conspicuity.

Image review & registration validation. Judge registrations for geometric accuracy AND downstream impact — good alignment drives accurate contour propagation, less on-table editing, and reliable motion management.

Decision point: proceed with adaptation vs. couch-shift-and-treat without adaptation. Sound judgment at these branch points is an essential competency.

CBCT-gART (Ethos)

Target must lie within the MLC-modifiable field, or the adaptive workflow cannot proceed.

MR-gART (Unity)

Patient setup and shift allows the adaptive treatment, scan length sufficient and within field length limit

BGRT

Coordinate with Nuclear Medicine: injection, uptake & imaging within the biologic time window.



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Contouring — clinically meaningful, fast, collaborative

- ▶ **Edit what changes dose or coverage**

Review propagated / auto-segmented OARs and targets; make clinically meaningful edits — not cosmetic ones that add time without impact.

- ▶ **Know how much accuracy actually matters**

Balance contour fidelity against on-couch efficiency; use contour “keys” / priorities defined at preplan to focus edits with the highest dosimetric value.

- ▶ **Judge the source — rigid, deformable, or AI**

Evaluate the fidelity of each generation method and decide when and how to intervene; some systems allow margin-expansion edits online.

- ▶ **Co-pilot with the physician**

Effective editing is collaborative — technical efficiency + clinical judgment to finish within the time window.

BGRT today

No online adaptive component — contouring remains offline, but the same principles of review, efficiency & clinical judgment apply.

Replan, approve, deliver — under the clock

Replanning & optimization. Comprehend robust objective strategies; read optimization behavior in real time; know when to accept a near-optimal plan vs. iterate; apply stopping criteria that protect session length.

Plan evaluation & approval. Confirm contours, margins, density overrides & dose metrics; validate with a secondary calc and a verification CBCT. **Final medical appropriateness remains the physician's accountability.**

Delivery & real-time monitoring. Confirm the beam-on anatomy truly matches the approved plan; coordinate imaging, gating & plan monitoring while watching patient stability and efficiency.

EXAMPLE of MOTION MANAGEMENT BY MODALITY

CBCT-gART

SGRT-guided breath-hold: Ensure reproducible breath-hold positioning using SGRT and verify accuracy with CBCT; intervene when breath-hold reproducibility criteria are not achieved..

MR-gART

Real-time cine-MRI gating — Define/validate gating windows and manage intrafraction motion during delivery, including decisions on gating interruption, intervention, or temporary gating disablement.

BGRT

Align delivery to biologic signal localization and tracer timing; monitor signal quality, decay, and motion; make treatment decisions when tracking criteria are not maintained.



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03

Training, Credentialing & Ethics

- Building judgment under time pressure — before and after go-live

Build judgment before the first patient

1

Structured observations & site visits

Shadow mature programs: quantify throughput, staffing models & failure-mode mitigations → convert findings into local SOPs.

2

Vendor software emulators

Reproduce the UI, optimization, image review & decision gates — stress-test handoffs & timing without consuming vault time.

3

Retrospective case replays

Controlled imports of de-identified prior datasets to replay decision points and compare strategies.

4

Phantom dry runs / table-top rehearsals

End-to-end sessions with phantoms or digital twins — imaging, registration, editing, optimization, QA & delivery under realistic timing.



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Credential to the task — privilege by condition

- ▶ **Platform-specific trainings**
MR safety, isotope handling, imaging protocol training...etc.
- ▶ **Document who may do what**
Privileges specify tasks and supervision level — and vary by case complexity and modality.
- ▶ **Same discipline, different privileges**
e.g., a physicist may run prostate CBCT-gART independently but need supervision for pancreas MR / BGRT.

AVAILABLE CREDENTIALING REFERENCES

- ✓ MR / BGRT: modality-appropriate safety training — TG-284 for MR; RAM-license training for PET isotopes (see book chapter)
- TG-395 CT and X-ray guided oART
- ✓ ≥ 1 mock adaptive case on phantom or emulator before live treatment
- ✓ ≥ 1 supervised live case before working independently
- ✓ Contouring proficiency on relevant OARs / targets, evaluated by the Oversight Committee

Competency is perishable

ART workflows evolve rapidly—expect staffing needs and workflow metrics to change with platform upgrades.

Stay active

Define minimum case involvement — e.g., monthly or quarterly activity to remain current, with annual refreshers.

Partner with vendors

Couple refreshers with vendor-led sessions; record completion in local credentialing files.

Refresh after change

Mandatory refreshers after workflow changes, major software / hardware upgrades, or new imaging / planning features.

Write it into policy

Spell out participation thresholds, refresher schedules & documentation — for fairness, transparency & consistency.

Example- RTT Contour Training

- Contouring is not a routine component of traditional U.S. RTT curricula
- Core Knowledge Areas
 - Anatomy
 - ASRT online cross section anatomy
 - Contouring Fundamentals
 - Online free resources - eContour, ProKnow...etc.
 - Adaptive Therapy Workflow
 - Site-specific contour keys
 - Platform-specific tool
 - Decision making keys

**More to come- AAPM-COMP
2026 Annual Meeting**



ProKnow

eContour

Adaptive Therapist Training



International Journal of Radiation
Oncology*Biography*Physics
Available online 1 April 2025
In Press, Journal Pre-proof ? What's this?



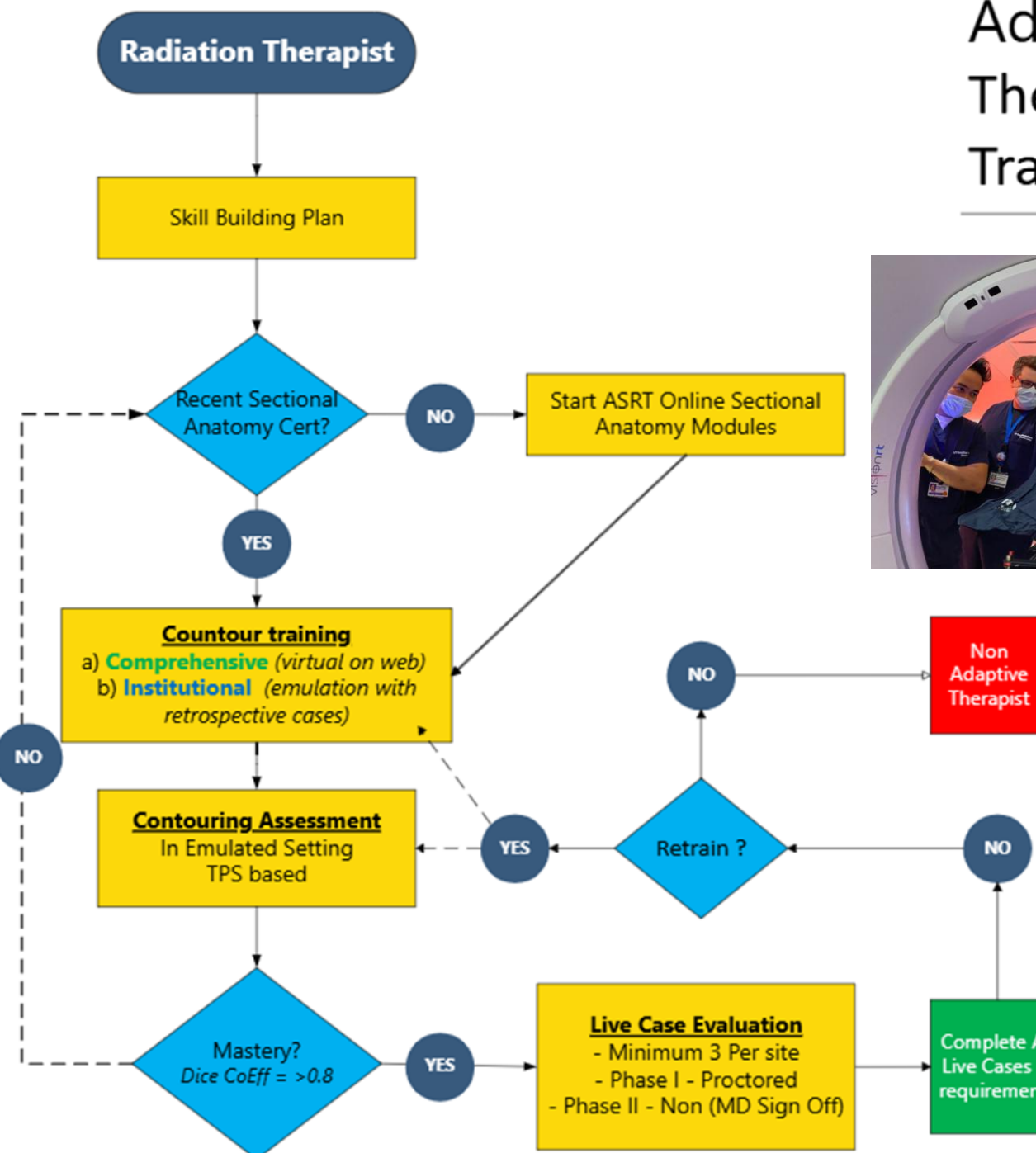
Physics Contribution

Feasibility and Impact of an RTT-Driven Workflow for CBCT-Guided Partial Breast Adaptive Radiotherapy

David Parsons , Sean Domal, Eric Chambers, Denise Salazar, Justin Visak, Mona Arbab, Francis Holgado, Dillon Li, Chinasa Okoro, Narine Wandrey, Zohaib Iqbal, Prasanna Alluri, Bin Cai, Hannah Keen, Jennifer Cleaton, Andrew Godley, David Sher, Shahed Badiyan, Asal Rahimi, Mu-Han Lin



RTT- vs Physician-Driven ART



Contour	Metric	Mean [%]	Standard Deviation [%]
Tumour Bed	V _{100%}	-0.004	0.204
	V _{95%}	-0.093	0.425
CTV	V _{100%}	-0.311	0.887
	V _{95%}	-0.572	1.305
PTV	V _{100%}	-0.948	2.168
	V _{95%}	-0.572	1.305

Metric	Contour	Mean	Standard Deviation
Dice Coefficient	Tumour Bed	0.96	0.06
	CTV	0.98	0.03
	PTV	0.98	0.02
Hausdorff Distance	Tumour Bed	0.62 mm	0.68 mm
	CTV	0.71 mm	0.71 mm
	PTV	0.73 mm	0.73 mm

Ethics under time pressure

Privacy & data security

Safeguard patient data within fast, data-rich adaptive workflows.

Transparent informed consent

Be clear about adaptive workflows and any additional imaging.

Disclose conflicts of interest

Especially when working with vendors and new technologies.

Equity of access

Promote fair patient access to adaptive therapy.

Professional conduct

Interdisciplinary respect, timely communication & fair decisions under pressure.

Make it explicit: embed ethics in training, credentialing & periodic refreshers — with formal acknowledgement by each team member.



04

Staffing & Sustainability

- Role-flexible models, the hidden workload, and where ART is heading

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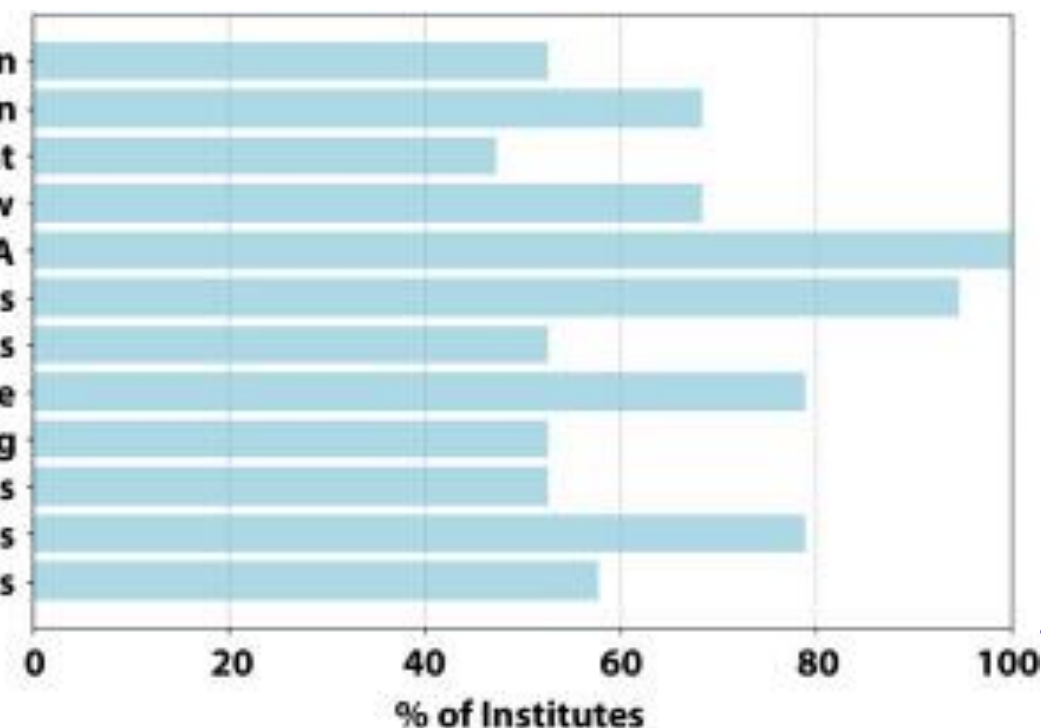
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^aDepartment of Radiation Oncology, UT Southwestern Medical Center, Dallas, Texas; and ^bDepartment of Radiation Medicine & Applied Sciences, UC San Diego Health, San Diego, California

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Pre-Treatment: Patient Selection for Adaptation
Pre-Treatment: Ethos Planning Template Generation & Validation
Pre-Treatment: Initial Planning for a Specific Patient
Pre-Treatment: Initial Plan Review prior to MD Review
Pre-Treatment: Final Initial Plan Review & QA
On-Treatment: Cover Adaptive Treatments
Post-Treatment: Adaptive Fraction Review of Contours
Post-Treatment: Adaptive Fraction Review of Dose
Post-Treatment: Adaptive Dose Tracking or Trending
Training/Credentialing of Adaptive Therapists
Training/Credentialing of Adaptive Dosimetrists
Training/Credentialing of MDs for Online Adaptive Process



BRIEF REPORT

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Online Adaptive Radiation Therapy: Clinical
Insights From a Nationwide Staffing Survey**

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North America Survey

Domain	CBCT-gART (Ethos Survey, n=19)	MRgRT Survey (n=22), Manuscript in Prep
Program Maturity	Most institutions treat 1-2 adaptive patients/day/machine	Median 6 adaptive fractions/day/machine
On-couch Coverage Model	No dominant national model reported; shared responsibility with IGRT physicist of the day is common	68% use Physicist-of-the-Day coverage
Reference Planning Staffing	Heavy physics involvement in adaptive planning and oversight	12/22 sites use dosimetry-led planning + physics review
Cross-Training	Not reported	Median 50% of physicists trained to create reference plans; several sites report all MRgRT physicists are plan-capable physicists
Role of Dosimetry	Limited online participation; mainly reference planning support* No Remote oART workflow	5/22 sites involve dosimetry online * Remote oART workflow available
RTT online Contour	50% institution has RTT involved in contour editing	50% of sites allow therapist contouring for selected sites
FTE Growth	>5 ART cases/day typically requires ≥1 additional physicist FTE	Most sites operate with ~1 console physicist FTE/machine

ART physics is bigger than the on-couch session

Programmatic

Commission adaptive workflows; QA units & TPS; validate imaging sequences; develop planning & treatment guidelines.

Hidden & ongoing

Training & credentialing; competency maintenance; emulator prep & troubleshooting; iterative process improvement.

Patient-level

Intent review; pre-treatment structure / objective checks; real-time on-couch support; post-treatment chart review.

Protected time

Dedicated effort to build & iterate adaptive guidelines and integrate upgrades — or it slips to after-hours.

Three Aspects Not Commonly Considered in Physics Staffing Models

- Reference planning & plan robustness review
 - Ref planning load can be calculated
 - Reference plan check/upstream plan review
 - 15 min-20 min ensure plan robustness
- On-couch coverage staffing
 - Can be estimated based on # of adaptive per day to decide on shared responsibility or dedicated physicist of the day
- Training, Credentialing, and Perishable Skill Maintenance
 - Oftentime underestimated

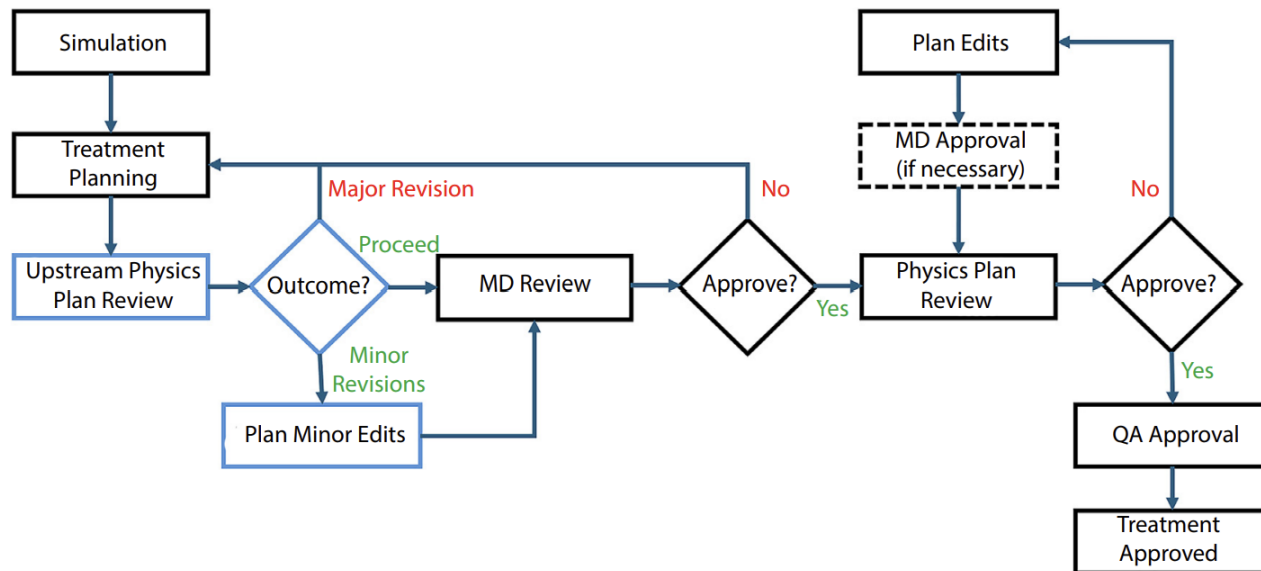
Upstream Plan Check

INTERNATIONAL JOURNAL OF
RADIATION ONCOLOGY • BIOLOGY • PHYSICS
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PHYSICS CONTRIBUTION

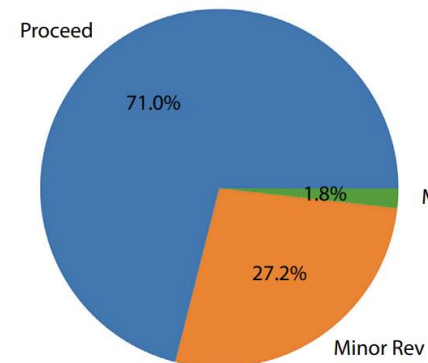
Establishing a Safety Net in X-Ray-Based Online Adaptive Radiation Therapy: Early Detection of Planning Deficiencies Through Upstream Physics Plan Review

Zohaib Iqbal, PhD, Yang Kyun Park, PhD, David Parsons, PhD, Andrew Godley, PhD, Steve Jiang, PhD, David Sher, MD, Shahed Badiyan, MD, Robert Timmerman, MD, and Mu-Han Lin, PhD

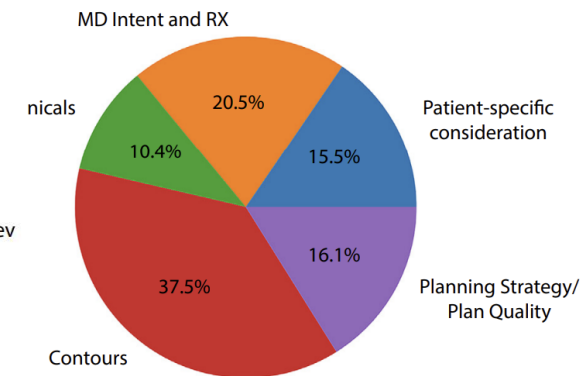


2024, Ethos

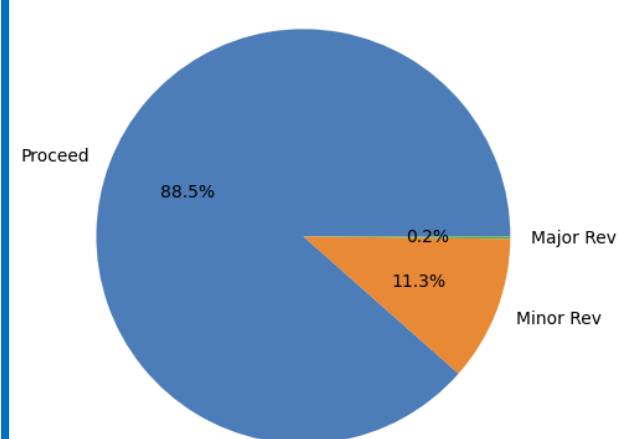
Upstream Plan Review Results



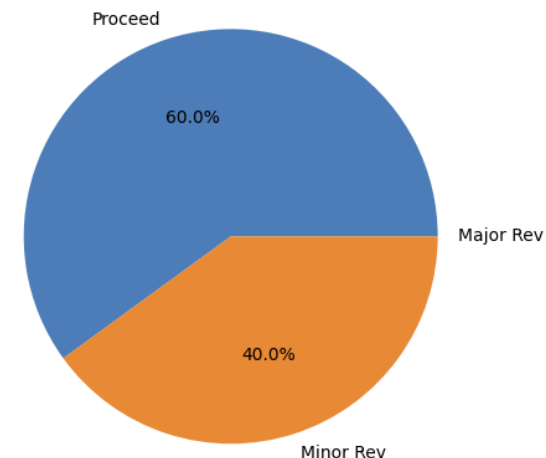
Distribution of Issues Identified during Plan Review



Upstream Plan Review Results Ethos 2026



Unity 2026



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Practical Approach to Adaptive Therapy Staffing Calculation

- 1. Define Clinical Scope
 - Treatment sites
 - Adaptive frequency (daily, weekly, selective)
 - Expected patient volume
- 2. Design On-Couch Capacity
 - Number of adaptive slots/day
 - Consistent scheduling simplifies staffing
 - Example: 2 adaptive slots/day (8:30 AM & 10:00 AM)
- 3. Translate Slots into Planning Workload
 - Daily adapt (20 fx/patient) → ~2 reference plans/month
 - Weekly adapt → ~10 reference plans/month
 - On-couch capacity drives upstream planning demand



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Practical Approach to Adaptive Therapy Staffing Calculation

- 4. Build the Coverage Model
 - MD / Physicist / Dosimetrist / RTT
 - ART physicist of the day vs Shared responsibility?
- 5. Include Often-Forgotten Effort
 - Commissioning & implementation
 - Training & credentialing
 - New staff onboarding
 - Software upgrade retraining
 - Perishable skill maintenance

Practical Radiation Oncology® (2025) 15, e72–e81



Basic Original Report

Quantitative Assessment of Full-Time Equivalent Effort for Kilovoltage-Cone Beam Computed Tomography Guided Online Adaptive Radiation Therapy for Medical Physicists

Natalie N. Viscariello, PhD,^{a,*} Kristen McConnell, PhD, MBA,^{b,c} Joseph Harms, PhD,^a Joel A. Pogue, PhD,^a Xenia Ray, PhD,^d Eric Laugeman, PhD,^e Richard A. Popple, PhD,^a Dennis N. Stanley, PhD,^{a,1} and Carlos E. Cardenas, PhD^{a,1}

^aDepartment of Radiation Oncology, University of Alabama at Birmingham, Birmingham, Alabama; ^bDepartment of Radiation Oncology, Miami Cancer Institute, Baptist Health South Florida, Miami, Florida; ^cDepartment of Radiation Oncology, Herbert Wertheim College of Medicine, Florida International University, Miami, Florida; ^dDepartment of Radiation Oncology, University of California San Diego, San Diego, California; and ^eDepartment of Radiation Oncology, Washington University St. Louis, St. Louis, Missouri



Where ART is heading

Lower the barriers to hands-on practice — shared infrastructure accelerates onboarding and lets community practices adopt ART.

Competency-driven teams

Role-flexible, top-of-training practice — not rigid, title-based hierarchies.

Shared training ecosystems

Case libraries, cloud emulators / sandboxes & standardized, role-mapped curricula.

Cross-modality training

Dual-trained staff for MR safety / distortion and PET tracer handling & timing.

AI & automation

Augment contouring, objective selection & QA — with human oversight, never replacement.

Minimum-expertise consensus

Standards anchored on competencies & credentialing, portable across sites & modalities.

IN SUMMARY

Five things to take home

1

Credential to the task, not the title

Organize ART by the work and skillset; let competent, credentialed staff practice at the top of their training.

2

Structure makes ART safe & scalable

Oversight committee + working groups + huddles + incident learning = a coordinated system of care.

3

Mind the chain & the modality

Early quality (registration!) ripples downstream; CBCT, MR & BGRT each add specific demands.

4

Build judgment before go-live

Site visits, emulators, case replays & phantom dry runs; then maintain competency deliberately.

5

Protect the physics effort

The hidden workload is real — document workflows, plan protected time, and advocate for FTEs.



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of PHYSICISTS IN MEDICINE



2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

JUNE 16–20 | UNIVERSITY OF MICHIGAN

Design your team and workflow around the work itself — not job titles.

DESIGNING ON-COUCH COVERAGE

♥ — is like — ♥

BUILD YOUR OWN BEAR!

The Right Skills.
The Right Team.
The Right Coverage.

STEP 1: BASED ON YOUR PLATFORM, LIST OUT THE SKILL SETS NEEDED

CBCT-gART (e.g., Varian Ethos)



- ✓ Image review & registration
- ✓ Contouring & clinical judgment
- ✓ Adaptive planning (automated)
- ✓ Plan evaluation & approval
- ✓ Delivery & motion management
- ✓ QA & documentation
- ✓ Communication & escalation

MR-gART (e.g., Elekta Unity)



- ✓ MR image review & registration
- ✓ MR safety & patient screening
- ✓ Contouring & clinical judgment
- ✓ Adaptive planning (objectives editable)
- ✓ Plan evaluation & approval
- ✓ Delivery & motion management
- ✓ MR-specific QA & physics checks
- ✓ Communication & escalation

BGRt (PET-Linac) (Offline Today)



- ✓ PET image review & interpretation
- ✓ Tracer timing & uptake window management
- ✓ Contouring & clinical judgment
- ✓ Offline adaptive planning
- ✓ Plan evaluation & approval
- ✓ QA & documentation
- ✓ Nuclear medicine coordination & communication

Start with the work. Add the skills. Build the model that fits your clinic.



EVERY DEPARTMENT BUILDS DIFFERENTLY.
The goal isn't the same bear — it's the same standard of safe, high-quality adaptive care.

STEP 2: MIX, MATCH, AND ASSIGN (WHO DOES WHAT)



Physician

Clinical judgment
Approve plan
Escalation decisions



Physicist

Imaging & registration
Plan evaluation
Physics QA
Workflow oversight



Dosimetrist

Contouring
Plan QA
Plan integrity checks



Therapist

Patient setup
Delivery
Motion mgmt
On-console workflows



Cross-trained team members

Back-up roles
Task flexibility
Coverage continuity



Credential to the task, not the title.

STEP 3: DESIGN PROPER TRAINING / CREDENTIALING FOR YOUR TEAM

EXTRA LEARNING NEEDED (BEYOND FOUNDATIONAL)

- Platform-specific knowledge (CBCT / MR / PET-Linac)
- Adaptive workflow & decision-making
- Imaging interpretation skills
- Plan evaluation & adaptive QA
- Safety, ethics & escalation protocols
- Communication & teamwork in real-time

TIMELINE FOR TRAINING (EXAMPLE)

- Orientation (1–2 weeks): Didactics, workflow overview, safety & policies
- Supervised training (4–8 weeks): Hands-on with preceptors
- Competency assessment (1–2 weeks)
- Provisional practice (2–4 weeks)
- Independent practice (after successful credentialing)

RESOURCES FOR TRAINING

- Vendor training programs & e-learning modules
- In-house didactics & case reviews
- Simulation / dry-run sessions
- Checklists, job aids & SOPs
- Peer observation & feedback
- Professional courses, webinars & conferences

CREDENTIALING APPROACH

- Education requirements
- Observed cases with checklist
- Written / online assessment
- Direct observation of performance
- Ongoing competency & re-credentialing



Train to the task. Prove the skill. Reassess regularly.

Use your platform strengths. Build on your team's strengths. Close your gaps intentionally.



DIFFERENT SKILLS. DIFFERENT TEAMS. ONE GOAL:
SAFE, CONSISTENT, HIGH-QUALITY ADAPTIVE CARE — EVERY DAY.

Continuous learning + good communication = better adaptive care for every patient.



“We adapt well — every day.”

Build scalable, role-flexible ART teams grounded in clear governance, shared competencies, and continuous learning.

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 - Kara James, RTT
- **Physicians**
- **Therapist Team**

Adaptive Therapy Physics Team



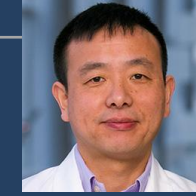
Mu-Han Lin, PhD



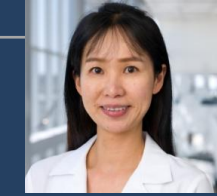
David Parsons, PhD



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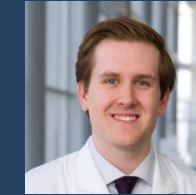
Weiguo Lu, PhD



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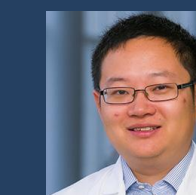
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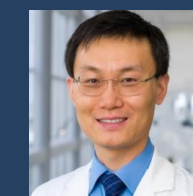
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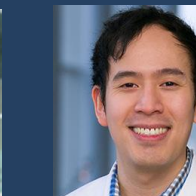
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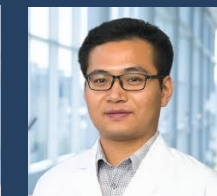
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